

BMET3997/9997

Physiological signal sensing and processing

Major project: Tips



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We acknowledge the tradition of
custodianship and law of the Country on which
the University of Sydney campuses stand.
We pay our respects to those who have cared
and continue to care for Country.



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QRS detector

- **Filter signal to remove baseline and high-frequency noise before QRS detection.**
- **For 3997 students: use *filtfilt* to ensure your filter has zero group delay (and therefore maintains shape of signal)**
- **If using a threshold, then make the threshold adaptive**
- **Try running your detector on the test data and plot the signals and detections points to confirm it is working satisfactorily**

HRV parameters

- **Develop your algorithms using the training data and the expert detections**
- **Confirm that your average HRV parameters are almost identical to the values provided in the instructions (especially for the time domain measures)**
- **Once working with the expert detections, try using your own QRS detections and check if any major differences.**



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